

Q: Find the generator of the cyclic group defined by the set $G = \langle \mathbb{Z}_6, + \rangle$

Ans $\mathbb{Z}_6 = \{0, 1, \dots, 5\}$

$$10 \bmod 6.$$

$$1 \bmod 6 = 0$$

$$1 \bmod 6 = 1$$

$$1^2 \bmod 6 = 2 \bmod 6 = 2.$$

$$1^3 \bmod 6 = 3 \bmod 6 = 3$$

$$1^4 \bmod 6 = 4 \bmod 6 = 4.$$

$$1^5 \bmod 6 = 5 \bmod 6 = 5.$$

$$1^6 \bmod 6 = 6 \bmod 6 = 0.$$

$$0^0 \bmod 6 = 0, 0^1 \bmod 6, 0^2 \bmod 6, 0^3 \bmod 6, 0^4 \bmod 6, 0^5 \bmod 6, 0^6 \bmod 6 = 0.$$

$$2^0 \bmod 6 = 1^0 \bmod 6 = 0.$$

$$2^1 \bmod 6 = 2^1 \bmod 6 = 2.$$

$$2^2 \bmod 6 = 4^1 \bmod 6 = 4$$

$$2^3 \bmod 6 = 6^1 \bmod 6 = 0$$

$$2^4 \bmod 6 = 8^1 \bmod 6 = 2.$$

$$2^5 \bmod 6 = 10^1 \bmod 6 = 4$$

$$2^6 \bmod 6 = 12^1 \bmod 6 = 0.$$

$$3^0 \bmod 6 = 1^0 \bmod 6 = 0.$$

$$3^1 \bmod 6 = 3^1 \bmod 6 = 3.$$

$$3^2 \bmod 6 = (3+3)^1 \bmod 6 = 0.$$

$$3^3 \bmod 6 = (3+3+3)^1 \bmod 6 = 3.$$

$$3^4 \bmod 6 = 12^1 \bmod 6 = 0.$$

$$3^5 \bmod 6 = 15^1 \bmod 6 = 3.$$

$$3^6 \bmod 6 = 18^1 \bmod 6 = 0$$

$$4^0 \bmod 6 = 1^0 \bmod 6 = 0.$$

$$4^1 \bmod 6 = 4^1 \bmod 6 = 4.$$

$$4^2 \bmod 6 = 8^1 \bmod 6 = 2.$$

$$4^3 \bmod 6 = 12^1 \bmod 6 = 0.$$

$$4^4 \bmod 6 = 16^1 \bmod 6 = 4.$$

$$4^5 \bmod 6 = 20^1 \bmod 6 = 2.$$

$$4^6 \bmod 6 = 24^1 \bmod 6 = 0.$$

$$5^0 \bmod 6 = 1^0 \bmod 6 = 0.$$

$$5^1 \bmod 6 = 5^1 \bmod 6 = 5$$

$$5^2 \bmod 6 = 10^1 \bmod 6 = 4$$

$$5^3 \bmod 6 = 15^1 \bmod 6 = 3.$$

$$5^4 \bmod 6 = 20^1 \bmod 6 = 2$$

$$5^5 \bmod 6 = 25^1 \bmod 6 = 1$$

$$5^6 \bmod 6 = 30^1 \bmod 6 = 0.$$

Thus the generators are 1 & 5.

MODULO PROPERTIES

$$1) (a+b) \bmod n = (a \bmod n + b \bmod n) \bmod n$$

$$2) (a-b) \bmod n = (a \bmod n - b \bmod n) \bmod n$$

$$3) ab \bmod n = (a \bmod n \times b \bmod n) \bmod n$$

$$Q: 11^{232} \bmod 13$$

$$1 = 2^0 + 2^1 + 2^2 = 1 + 2 + 4$$

$$\Rightarrow 11^{(1+2+4)} \bmod 13$$

$$\Rightarrow 11 \cdot 11^2 \cdot 11^4 \bmod 13$$

By property 3.

$$\Rightarrow (11 \bmod 13 \cdot 11^2 \bmod 13 \cdot 11^4 \bmod 13) \bmod 13$$

$$= [11 \bmod 13 \cdot 11^2 \bmod 13 \cdot 11^4 \bmod 13] \bmod 13$$

$$= 11 \cdot 4 \cdot 3 \bmod 13$$

$$= [11 \times 4 (11^2 \cdot 11^2) \bmod 13] \bmod 13$$

$$= [11 \times 4 (11^2 \bmod 13 \times 11^2 \bmod 13) \bmod 13] \bmod 13$$

$$= (11 \times 4 \times 4 \times 4 \bmod 13) \bmod 13$$

$$= (704 \bmod 13) \bmod 13$$

$$= 2 \bmod 13$$

$$= \underline{2}$$

$$Q: 11^{232} \bmod 13$$

$$Ans: 232 \Rightarrow 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6$$

$$= 127$$

$$232 - 127 = 105$$

$$105 \Rightarrow 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + 2^5 = 63$$

$$105 - 63 = 42$$

$$42 \Rightarrow 2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 31$$

$$42 - 31 = 11$$

$$11 = 2^0 + 2^1 + 2^3 =$$

$$(1+2+4+8+16+32+64) (1+2+4+8+16+32)$$

$$\times 11 (1+2+4+8+16) (1+2+4) \bmod 13$$

$$\left[\left(11 \cdot 11^2 \cdot 11^4 \right) \cdot 11^8 \cdot 11^{16} \cdot 11^{32} \cdot 11^{64} \right] \left(11 \cdot 11^2 \cdot 11^4 \cdot 11^8 \cdot 11^{16} \cdot 11^{32} \right) \bmod 13$$

$$\left(11 \times 4^3 \times 4^8 + 4^{16} + 4^{32} \right) \left(11 \times 4^2 + 4^4 \right)$$

$$11^{232} \bmod 13$$

$$= 11^{(8+32+64+128)} \bmod 13$$

$$= 11^8 \cdot 11^{32} \cdot 11^{64} \cdot 11^{128} \bmod 13$$

$$= (11^8 \bmod 13 \cdot 11^{32} \bmod 13 \times 11^{64} \bmod 13 \times$$

$$11^{128} \bmod 13) \bmod 13$$

$$11 \bmod 13 = 11$$

$$11^2 \bmod 13 = (11 \bmod 13 \cdot 11 \bmod 13) \bmod 13$$

$$= (121) \bmod 13$$

$$= \underline{4}$$

$$11^4 \bmod 13 = (11^2 \bmod 13 \cdot 11^2 \bmod 13) \bmod 13$$

$$= (4 \cdot 4) \bmod 13$$

$$= 3$$

$$11^8 \bmod 13 = (11^4 \bmod 13 \cdot 11^4 \bmod 13) \bmod 13$$

$$= (3 \cdot 3) \bmod 13 = 9 \bmod 13$$

$$= \underline{9}$$

$$11^{16} \bmod 13 = (11^8 \bmod 13 \cdot 11^8 \bmod 13) \bmod 13$$

$$= (9 \times 9) \bmod 13 = 81 \bmod 13$$

$$= \underline{3}$$

$$11^{32} \bmod 13 = (11^{16} \bmod 13 \cdot 11^{16} \bmod 13) \bmod 13$$

$$= (3 \cdot 3) \bmod 13 = 9 \bmod 13$$

$$= \underline{9}$$

$$11^{64} \bmod 13 = 3$$

$$11^{128} \bmod 13 = 9$$

$$\Rightarrow (9 \times 9 \times 3 \times 9) \bmod 13 = 2187 \bmod 13$$

$$= \underline{2}$$

EUCLIDEAN ALGORITHM:

- 1) GCD
- 2) Multiplicative Inverse.

(i) GCD:

eg) To find $\text{gcd}(a, b)$ Put a in x_1 b in x_2 while $x_2 > 0$.If $x_2 > 0$ Quotient, $q = x_1 / x_2$ andremainder, $r = x_1 - q x_2$ is

found.

then x_1 becomes new x_2 and x_2 becomes the obtained r ,and it continues until $x_2 = 0$.

Once greater than zero.

 $\text{GCD}(a, b) = x_1$.eg) $\text{gcd}(25, 60)$

q	x_1	x_2	r
	25	60	
2	60	25	10
2	25	10	5
2	10	5	0
	5	0	

 $\Rightarrow \text{GCD}(25, 60) = 5$.Q: $\text{GCD}(1970, 1066)$

q	x_1	x_2	r
	1970	1066	
1	1066	904	162
5	904	162	94
1	162	94	68
1	94	68	26
2	68	26	16
1	26	16	10
1	16	10	6
1	10	6	4
1	6	4	2
2	4	2	0
	2	0	

 $x_1 = 2$ $\Rightarrow \text{GCD}(1970, 1066) = 2$

Extended Euclidean Algorithm.

q	x_1	x_2	r	s_1	s_2	s	t_1	t_2	t
	1	0		1	0		0	1	

 $\Rightarrow \text{GCD}(1970, 1066)$ $s x a + t x b = \text{GCD}(a, b)$ $a = 12, b = 36$ $12x + 36y = \text{GCD}(12, 36)$ Take $(s_1, s_2) = (1, 0)$ $(t_1, t_2) = (0, 1)$ $s = s_1 - q s_2$ $t = t_1 - q t_2$ $r = x_1 - q x_2$ Q: Find $\text{GCD}(161, 28)$

$$\text{GCD}(a, b) =$$

$$S \times a + t \times b.$$

$$\Rightarrow -1 \times 161 + 6 \times 28$$

$$= -161 + 168$$

$$= f$$

Q: Find the multiplicative inverse of 11 in $\mathbb{Z}_{26}$.

n	q	n_1	n_2	L_1	L_2	L
1	2	26	11	0	1	-2
3	2	11	4	1	-2	5
1	1	4	3	0	5	-7
0	3	3	1	5	-7	26
		1	0	-7	26	

$$-7 \bmod 26 \Rightarrow -7 = 26 - 7 = 19$$

$$\Rightarrow 19 \pmod{26}$$

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Q: Find the multiplicative inverse

d. 12 in 726

q	x_1	x_2	x	t_1	t_2	t
2	26	12	2	0	1	-2
6	12	2	0	1	-2	13
	2	0		-2	13	

$$\Rightarrow -2 \pmod{26} \quad \cancel{\neq 12, 22, 12, 12}$$

~~24 mod 25~~ at $x_2 = 0, x_1 \neq 1$

~~= 244~~ Hence multiplicative

inverse does not exist

Q: Find the multiplicative inverse.

of 256 in 23334.

Value in t_1 will be the multiple^a
time inverse.

when $x_2 = 0$ & $x_1 = 1$ only
then multiplication inverse exists

q	q_1	q_2	q_3	L_1	L_2	t
13	3334	256	6	0	1	13
42	256	6	4	1	13	547
1	6	4	2	-13	547	560
2	4	2	0	547	560	1667
	2	0		-560	1667	

Ans 1) Operation: multiplication

$$Z_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$1 \bmod 10 = 1$$

as at $q_2 = 0$, $q_1 \neq 1$ therefore the multiplication inverse doesn't exist.

1) Q: Find the generators of the cyclic group defined by the set $G = \langle Z_{10}, \times \rangle$

✓ 2) Q: Find the multiplicative inverse of 29, 25, 27, and 32 in Z_{42} .

3) Q: Find the multiplicative inverse of 556 in Z_{6543} .

4) Q: $14^{323} \bmod 12$, $13^{513} \bmod 15$, $11^{134} \bmod 15$

5) Q: Using Euclidean algorithm, find the values of s and t and $\text{GCD}(3224, 6024)$.

6) Q: $\bmod 9$ multiplication table

$\times$
0
1
2
3
4
5
6
7
8

Ans 3) Inverse of 556 in Z_{6543}

q	q_1	q_2	q_3	L_1	L_2	t
11	6543	556	427	0	1	-11
1	556	427	129	1	11	12
3	427	129	+40	-11	12	-41
3	129	+40	129	12	-41	153
4	40	129	4	-47	153	-659
2	9	4	1	163	-659	1471
4	4	1	0	-659	1471	6543
1	1	0		1471	6543	

at $q_2 = 0$ $q_1 = 1$

Hence multiplicative inverse exists.

$\Rightarrow$ value of $L_1 = 1471$

~~1471~~ $\bmod 6543$

$\Rightarrow$ inverse = 1471

$$4) a) 14^{323} \bmod 12$$

$$= (14^2 + 64 + 256) \bmod 12$$

$$= (14^2 \cdot 14^2 \cdot 14^{64} \cdot 14^{256}) \bmod 12$$

$$= (14^1 \bmod 12 \cdot 14^2 \bmod 12 \cdot 14^{64} \bmod 12 \cdot 14^{256} \bmod 12) \bmod 12$$

$$14 \bmod 12 = 2$$

$$\begin{aligned}
 14^2 \bmod 12 &= 14 \cdot 14 \bmod 12 \\
 &= (14 \bmod 12 \cdot 14 \bmod 12) \bmod 12 \\
 &= (2 \cdot 2) \bmod 12 \\
 &= 4 \bmod 12 \\
 &= \underline{4}
 \end{aligned}$$

$$\begin{aligned}
 14^4 \bmod 12 &= (14^2 \cdot 14^2) \bmod 12 \\
 &= (14^2 \bmod 12 \cdot 14^2 \bmod 12) \bmod 12 \\
 &= (4 \cdot 4) \bmod 12 = 16 \bmod 12 \\
 &= \underline{4}
 \end{aligned}$$

$$\begin{aligned}
 14^8 \bmod 12 &= (14^4 \bmod 12 \cdot 14^4 \bmod 12) \bmod 12 \\
 &= (4 \cdot 4) \bmod 12 \\
 &= \underline{4}
 \end{aligned}$$

$$14^{16} \bmod 12 = 4$$

$$14^{32} \bmod 12 = 4$$

$$14^{64} \bmod 12 = 4$$

$$14^{256} \bmod 12 = 4$$

$$\begin{aligned}
 &\Rightarrow (2 \cdot 4 \cdot 4 \cdot 4) \bmod 12 \\
 &= 128 \bmod 12 \\
 &= \underline{8}
 \end{aligned}$$

Irreducible polynomial

polynomial represents prime no but not full in mod

1) $x+1$ 2 bit : 3 (11)

2) x^2+x+1 3 bit : 7 (111)

3) x^3+x+1 4 bit : 11 (1011)

x^3+x^2+1 4 bit : 13 (1101)

Q: $(x^5+x^2+x)(x^7+x^4+x^3+x^2+x)$ in $GF(2^8)$ with irreducible polynomial $x^8+x^4+x^3+x+1$

$$\begin{aligned}
 &\text{Ans } (x^5+x^2+x)(x^7+x^4+x^3+x^2+x) \\
 &\Rightarrow x^{12}+x^9+x^8+x^7+x^6+x^5+x^4+x^3+x^2+x
 \end{aligned}$$

$$\begin{aligned}
 &\Rightarrow x^{12}+2x^9+2x^8+x^7+2x^6+2x^5+2x^4+2x^3+x^2 \\
 &\quad +2x+1 \quad \text{--- ①}
 \end{aligned}$$

Now taking mod of ① using $x^8+x^4+x^3+x+1$

$GF(2^8) \Rightarrow$ coefficients must be 0 or 1

" $2^8 \Rightarrow 00000000$

because $GF(2^8)$ is extended

version of $GF(2)$ as $GF(2)$ has (0, 1).

$\Rightarrow$ Cancelling the terms having coefficients other than 0, 1

$$\Rightarrow \text{①} \Rightarrow x^{12}+x^7+x^2$$

$$\Rightarrow (x^{12}+x^7+x^2) \bmod (x^8+x^4+x^3+x+1)$$

$$\begin{aligned}
 &\frac{x^4-1}{x^8+x^4+x^3+x+1} \overline{) x^{12}+x^7+x^2} \\
 &\quad -(x^{12}+x^8+x^7+x^5+x^4)
 \end{aligned}$$

In this step make the coefficient in $GF(2^8)$ i.e. 0, 1

$$\Rightarrow (-x^8-x^5-x^4-x^2) \bmod 2$$

$$\begin{aligned}
 &\Rightarrow \begin{matrix} -1+2 & -1+2 & -1+2 & -1+2 \end{matrix} \\
 &= \begin{matrix} 1 & 1 & 1 & 1 \end{matrix}
 \end{aligned}$$

$$\begin{array}{r}
 2^4 + 1 \\
 \Rightarrow 2^8 + 2^4 + 2^3 + 2 + 1 \mid 2^{12} + 2^4 + 2^2 \\
 \quad - 2^{12} + 2^8 + 2^7 + 2^5 + 2^4 \\
 \quad \quad - 2^8 - 2^5 - 2^4 - 2^2 \\
 \Rightarrow 2^8 + 2^5 + 2^4 + 2^2 \\
 \quad + (2^8 + 2^4 + 2^3 + 2 + 1) \\
 \quad \quad \{ 2^5 - 2^3 + 2^2 - 2 - 1 \\
 \quad \quad \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \\
 \quad \quad \quad 1 \quad -1+2 \quad 1 \quad -1+2 \quad -1+2
 \end{array}$$

Include this to in
GF(2⁸) i.e

⇒ final ans

$$\Rightarrow \underline{2^5 + 2^3 + 2^2 + 2 + 1}$$

$$a) x_2 = 0, x_1 = 1$$

Thus inverse exist

⇒ value of t, is its inverse

$$\Rightarrow \underline{1 + 2 + 2^3}$$

Q: Find the inverse of (x²+1) in
GF(2⁴)

$$\text{Ans. Hw } t = t_1 \oplus q t_2 \quad t = t_1 - q t_2$$

x = remainder

q = quotient

q	x ₁	x ₂	x	t ₁	t ₂	t
2 ² +1	2 ⁴ +2+1	2 ² +1	x	0	1	2 ² -1
x	2 ² +1	x	1	1	2 ² -1	1+2+2 ²
x	x	1	0	2 ² -1	1+2+2 ²	2 ⁴ -2 ²
	2+1	0		1+2+2 ²	2 ⁴ -2 ²	-2-1

$$\begin{array}{r}
 2^2 + 1 \\
 2^2 + 1 \mid 2^4 + 2 + 1 \\
 \quad - 2^4 + 2^2 \\
 \quad \quad 2^2 + 2 + 1 \\
 \quad \quad - 2^2 + 1 \\
 \quad \quad \quad 2
 \end{array}$$

$$\begin{array}{r}
 x \\
 x \mid 2^2 + 1 \\
 \quad - 2^2 \\
 \quad \quad 1
 \end{array}$$